

Module 9

The Four Dimensions of Product and Service Management

Syllabus

- 2.1 Introduction to the ITIL Four Dimensions of Product and Service Management (all Assessment Criteria included)
- 2.2 Internal factors and External factors (all Assessment Criteria included)
- 6.1 Introduction to AI (all Assessment Criteria included)
- 6.2 ITIL AI Governance (all Assessment Criteria included)



What you will learn



By the end of this module, you will be able to:

- explain the ITIL Four Dimensions of Product and Service Management and their importance in a holistic approach
- understand how AI can assist in the product and service lifecycle
- understand the ITIL AI Capability Model.

A holistic approach to service management



To support a holistic approach to digital product and service management, ITIL defines four dimensions that collectively are critical to the **effective** and **efficient** facilitation of **value** for **customers** and other **stakeholders** in the form of products and services.

These are:



**Organizations
and people**



**Information and
technology**



**Partners and
suppliers**



**Value streams
and processes**

The ITIL Four Dimensions of Product and Service Management

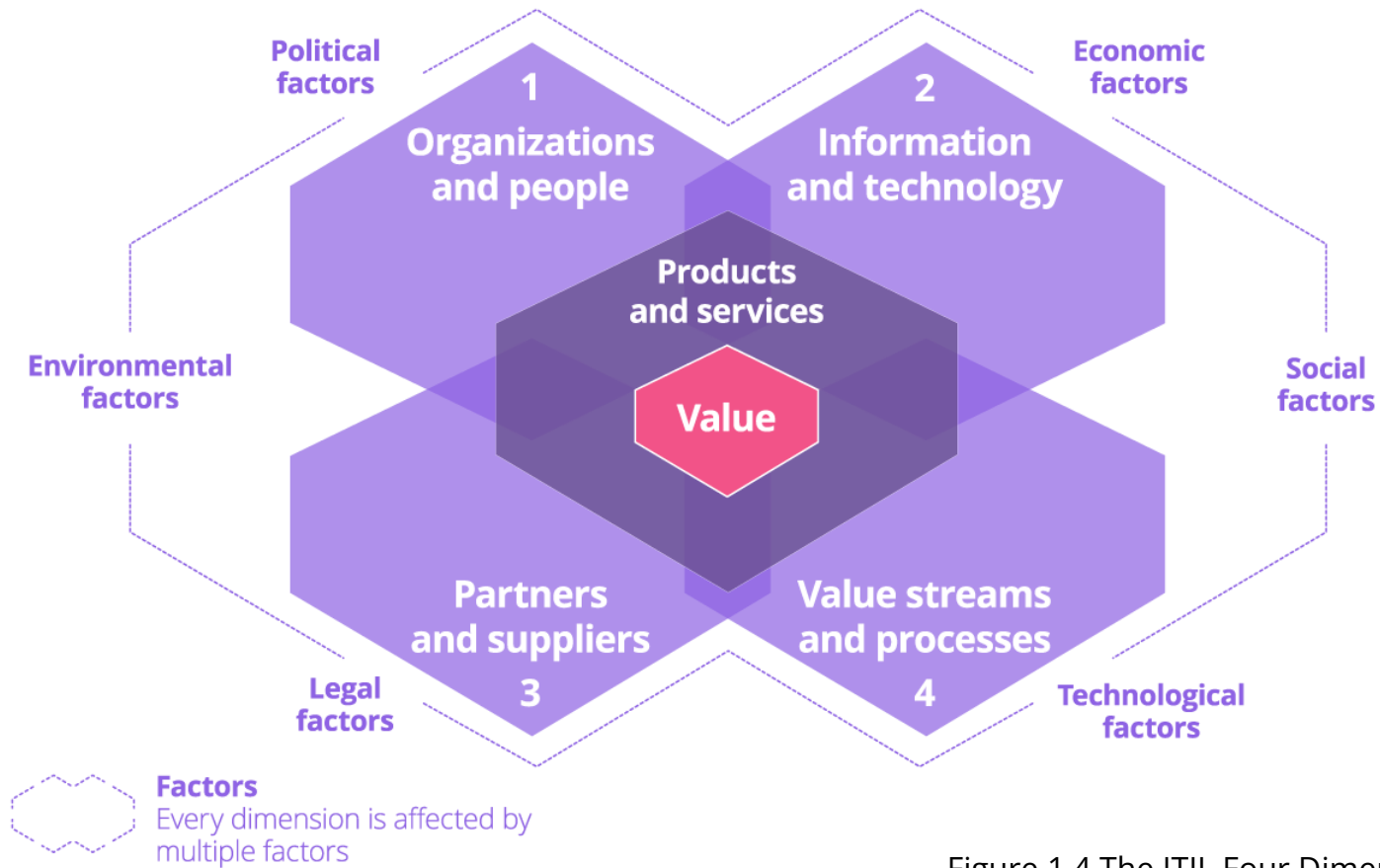
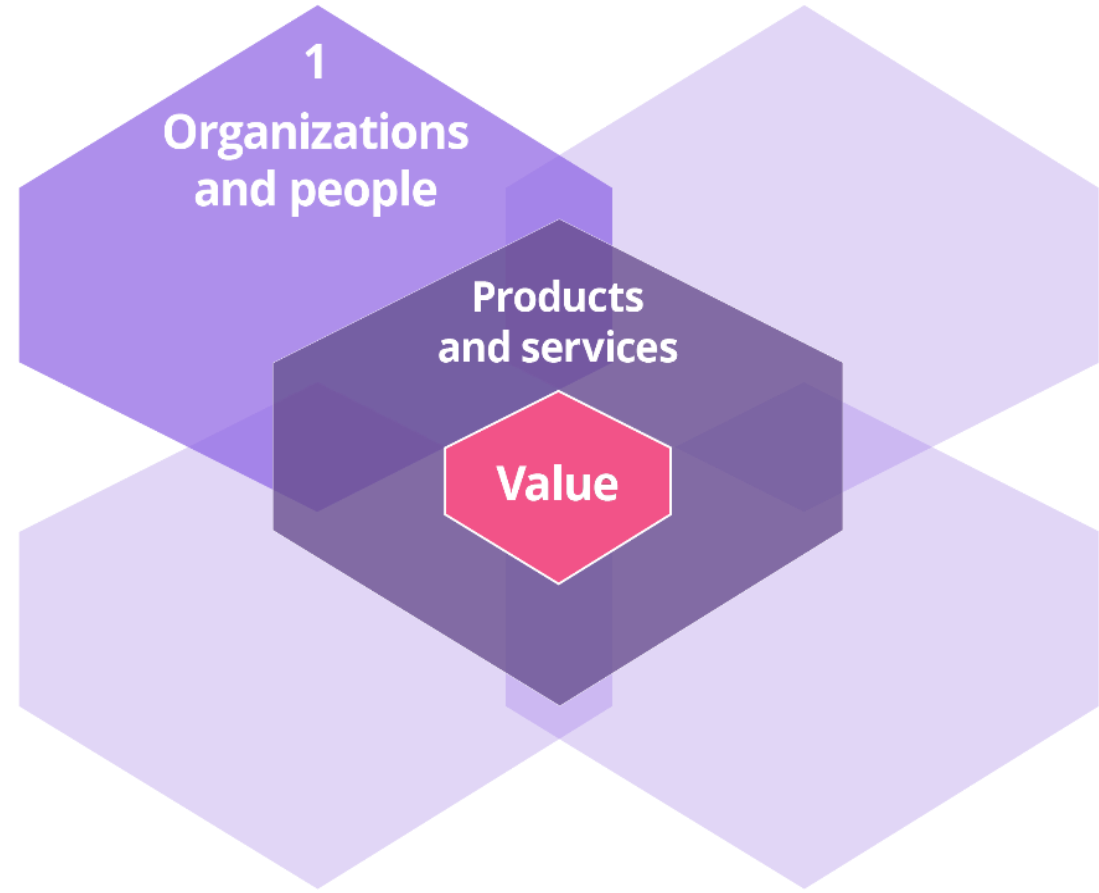
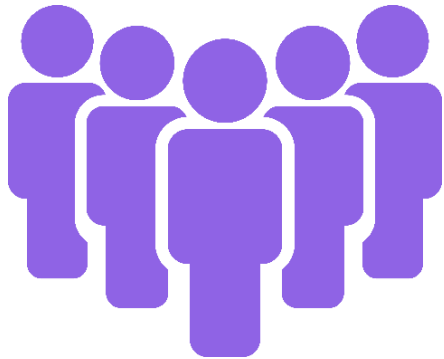


Figure 1.4 The ITIL Four Dimensions of Product and Service Management

Organizations and people

The complexity of organizations is growing. It is vital that their structure, roles, responsibilities, and systems of authority and communication are well-defined and support their overall strategy and the operating model.



Adapted from Figure 1.4 The ITIL Four Dimensions of Product and Service Management

Culture

How an organization carries out its work shapes the shared values and attitudes that, over time, become its **culture**.

Culture

A set of values shared by a group of people, expressed through people's behaviour and their ideas, beliefs, and practices.



Safety culture

An organizational culture in which people are comfortable being (and expressing) themselves.



Leadership

Leaders play an important role in shaping and maintaining organizational culture.

- **Effective leaders:**
 - **engage, inspire,** and **motivate** with integrity
 - focus on **value** and align with the overall vision
 - **create a healthy culture** by demonstrating the organizational values
 - **create and communicate** a clear vision
 - **commit** to knowledge and continual learning.

Good leadership balances the global good with individual self-interest.



Organizational structure and design

There are many organizational forms that are designed to optimize how people are positioned and managed to ensure optimal performance and strategic alignment.

Examples of how to categorize:

- Geography and location (centralized or distributed)
- Governance drivers (hierarchical, community-based, market-like)
- Scope of work and responsibility (functional, product/service-based, project-focused).

Core elements of any structure:

- Chain of command
- Roles and responsibilities
- Scope of control
- Decision-making authority
- Departments/teams



Conway's law and digital design

Definition

Conway's Law

"Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations".

Meaning

The way an organization communicates and is structured strongly influences how its systems, products, and services are designed.

Modern application

Reverse Conway Approach

Organizations deliberately design team structures to match the desired product or service architecture, enabling modular systems and end-to-end team ownership.

Skills and competencies

Organizations must continually develop and improve both technical and universal skills to meet evolving needs. Some important skills and competencies include:

- Creativity
- Attention to detail
- Problem-solving
- Critical thinking
- Teamwork / collaboration
- Interpersonal skills
- Work ethic
- Emotional intelligence
- Communication
- Adaptability / flexibility
- Time management
- Leadership
- Stress management



Organizations, people and AI

Humans excel at:

- creativity
- empathy
- contextual understanding.

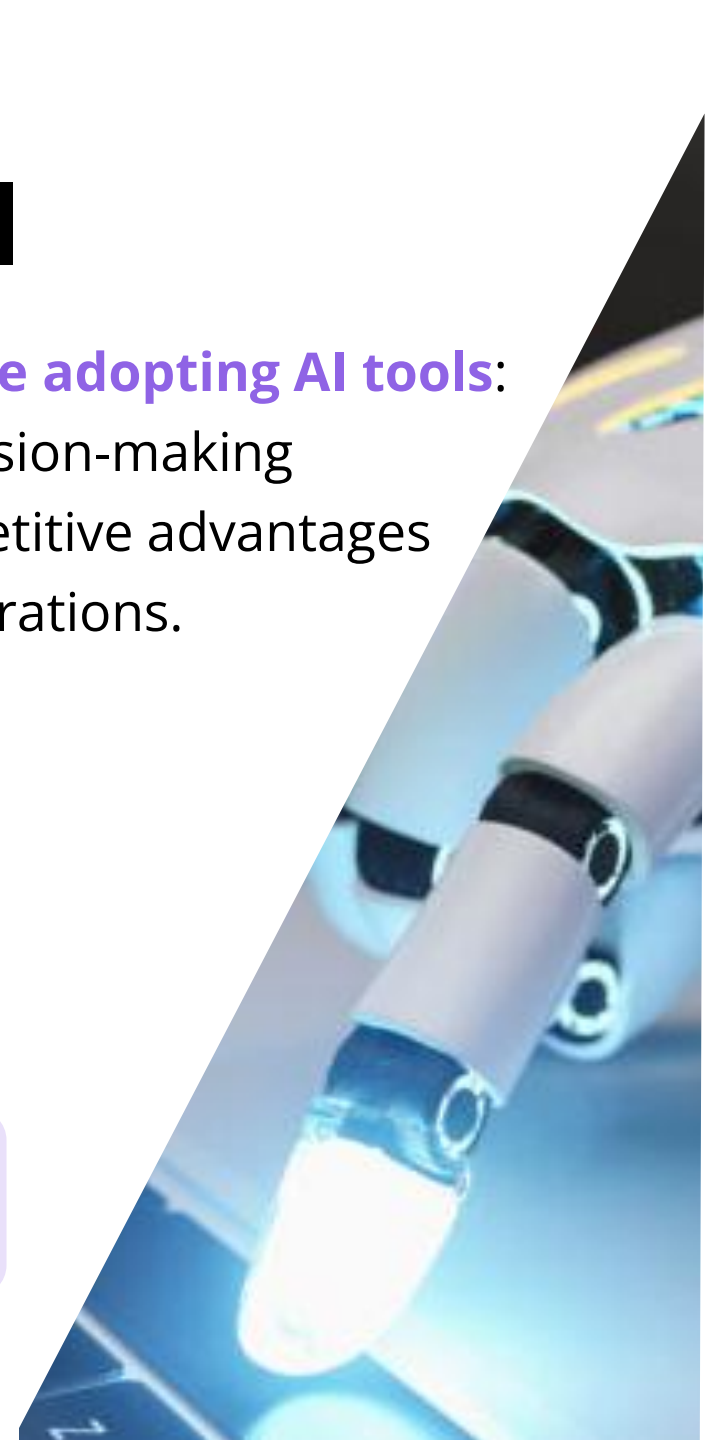
AI supports by:

- processing information: handle data quickly.
- recognizing patterns: detect trends, incidents, performance.
- reducing bias: support fairer, objective decisions.

Organizations are adopting AI tools:

- for critical decision-making
- creating competitive advantages
- optimizing operations.

AI is a collaborative partner, enhancing human strengths and offsetting limitations.



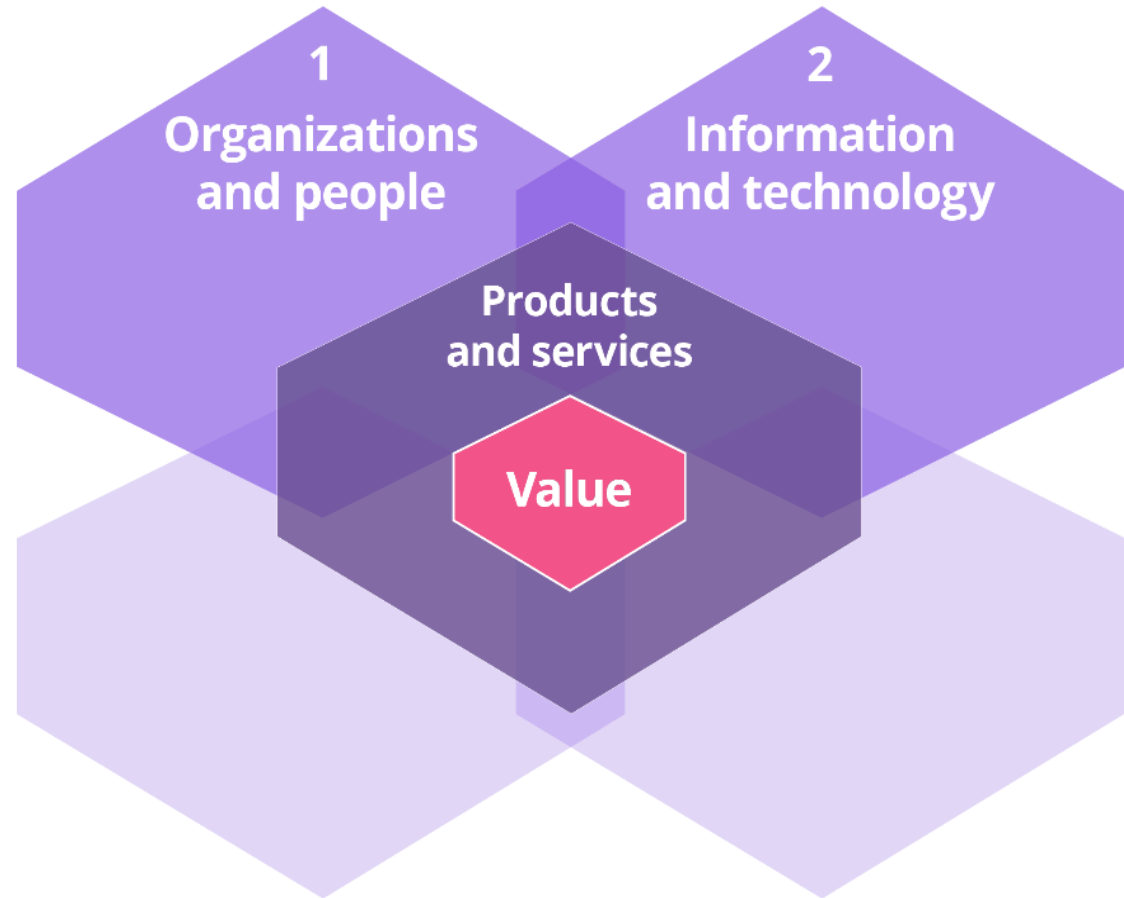
ITIL Car Rental's organization and people



Max: We have decided to revisit our organizational structure. We have several key products for both internal and external customers and we have made the decision to establish dedicated product teams to work on these offerings continuously. We understand that every product needs ongoing development and requires the right guidance and expertise, which is why we form permanent, product-focused teams.

Information and technology

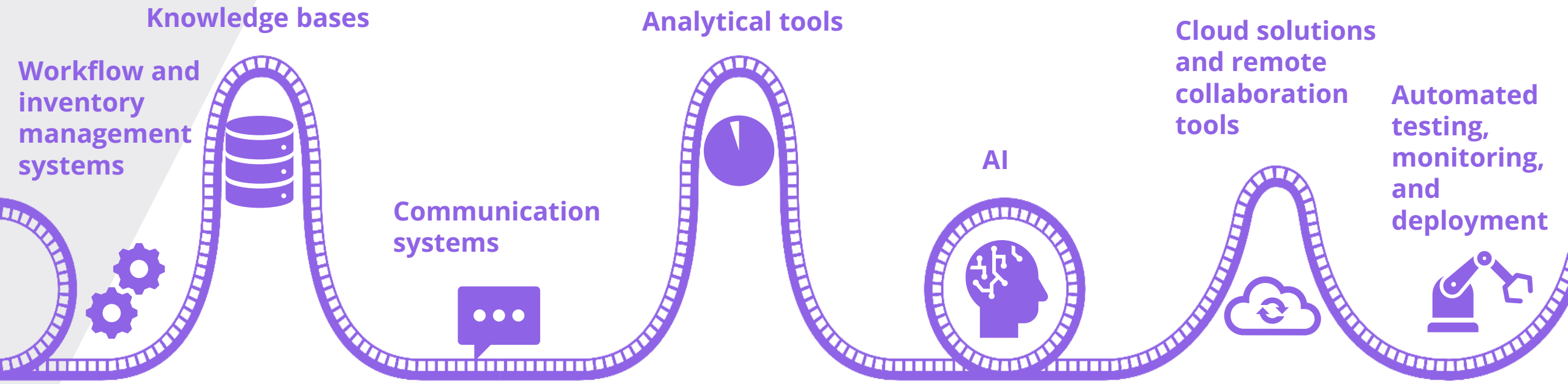
The 'information and technology' dimension addresses **data, information, and technologies** used in digital products and services, as well as those used as part of the organization's **product and service management systems**.



Adapted from Figure 1.4 The ITIL Four Dimensions of Product and Service management

Technology

The technologies that **support** product and service management include, but are not limited to:



Artificial Intelligence

AI supports digital product and service management by enabling smarter decisions and automation.

- AI enables systems to **learn, adapt,** and **execute** tasks that traditionally require human involvement.
- **Enhances** human **decision-making** and management.
- The 6C model describes AI's key functional capabilities.

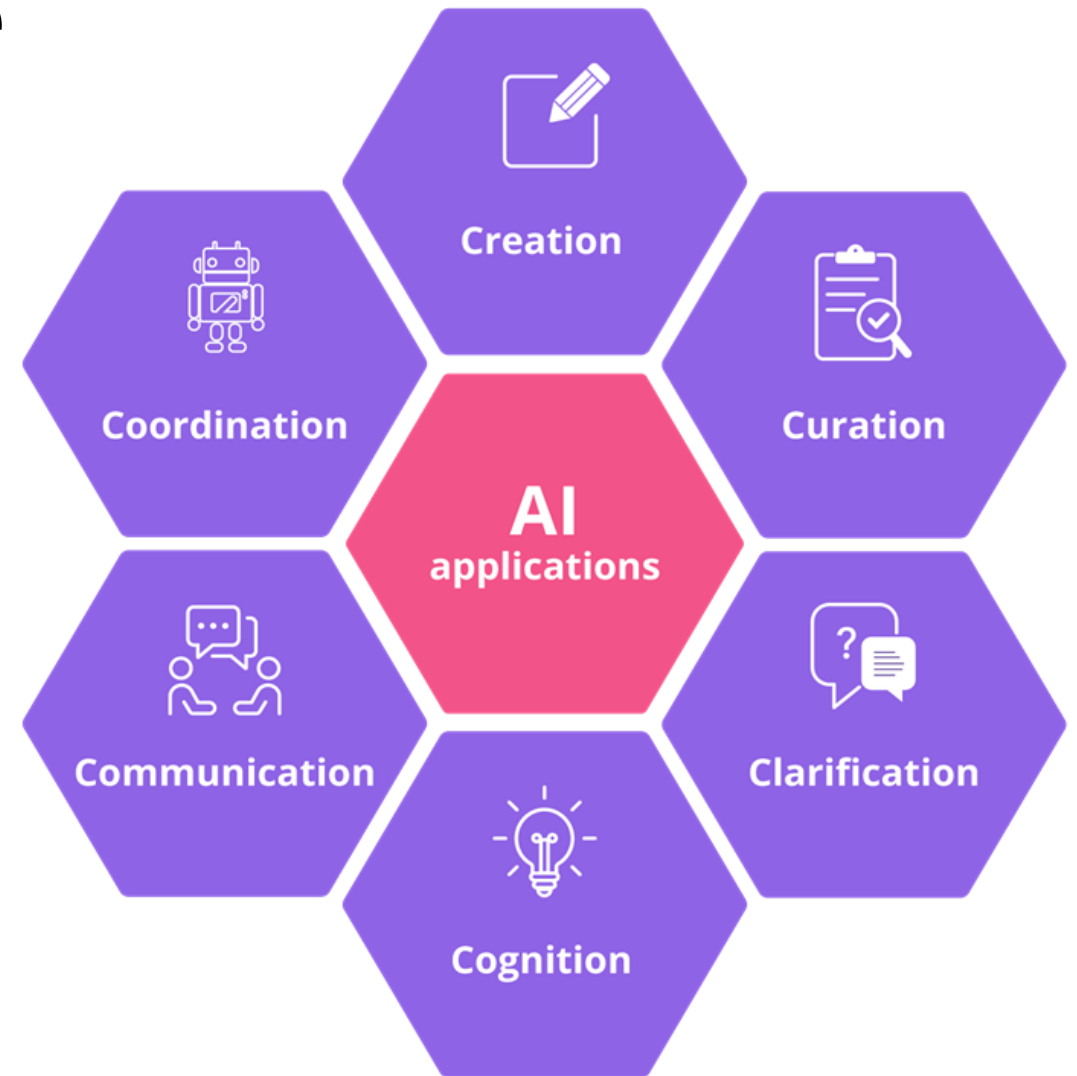


Figure 2.2 The ITIL AI Capability Model

AI challenges and governance



Governance

The system by which an organization directs, controls, and monitors the use of artificial intelligence to ensure that AI-enabled digital products and services create value while remaining ethical, secure, compliant, and aligned with organizational objectives.

With the **growth of GenAI**, organizations must also **address challenges** in governance, ethics, and compliance.



AI tools are **only as good as the data** they are trained on.



Data quality and control are **critical** for effective outcomes.



Governance **requirements** are **increasing** as regulations continue to evolve.



Generative AI adds **new opportunities** but **raises** new data-related **risks**.

Data, information, and knowledge

The amount of data generated and processed continues to grow daily.

- **Data:** raw facts and numbers, no intrinsic meaning.
- **Information:** processed data with context and meaning.
- **Knowledge:** applied information and experience.



Information has value only when it enables outcomes; out-of-context data has no value.

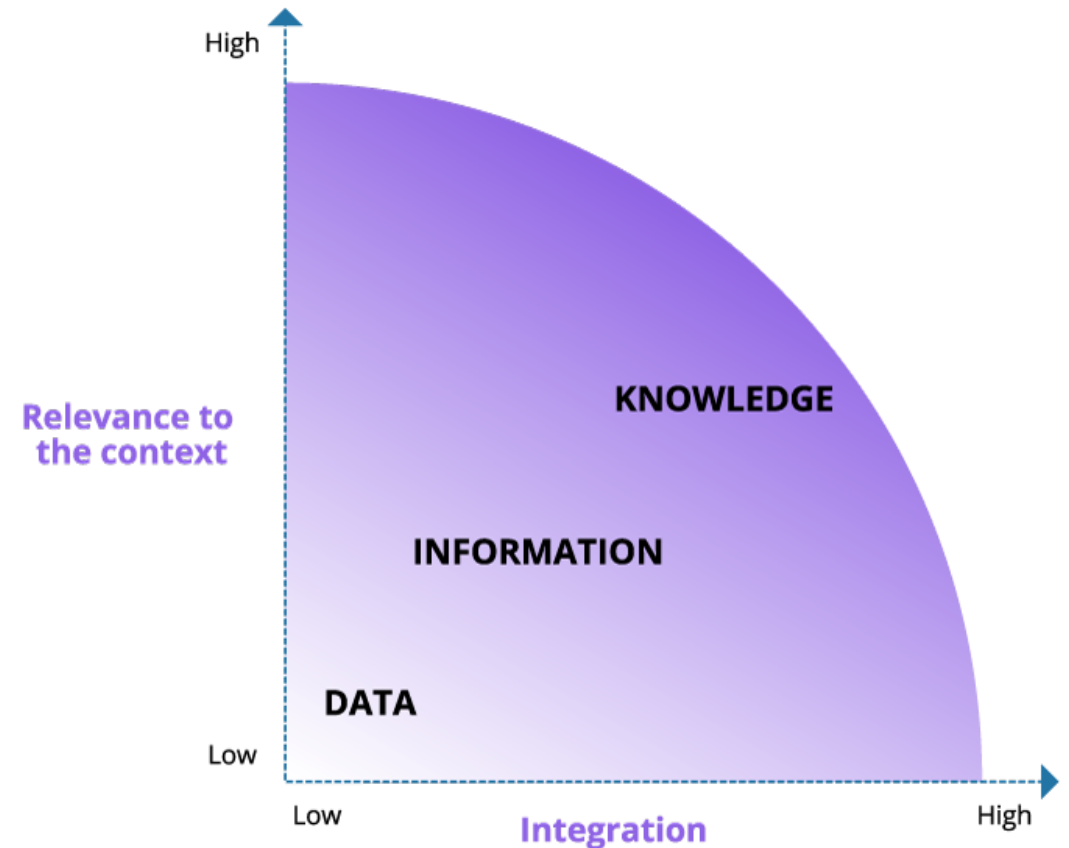


Figure 2.3 The unclear boundaries between data, information, and knowledge

Data governance



Data governance

A system of rules, policies, standards, processes, and controls organizations implement to manage their data assets effectively.

It ensures data remains secure, usable, and reliable throughout its lifecycle, built on six key principles:

- strategic alignment
- accountability and ownership
- policies & procedures
- transparency & auditability
- integrity
- adaptability.



Ensuring value from technology

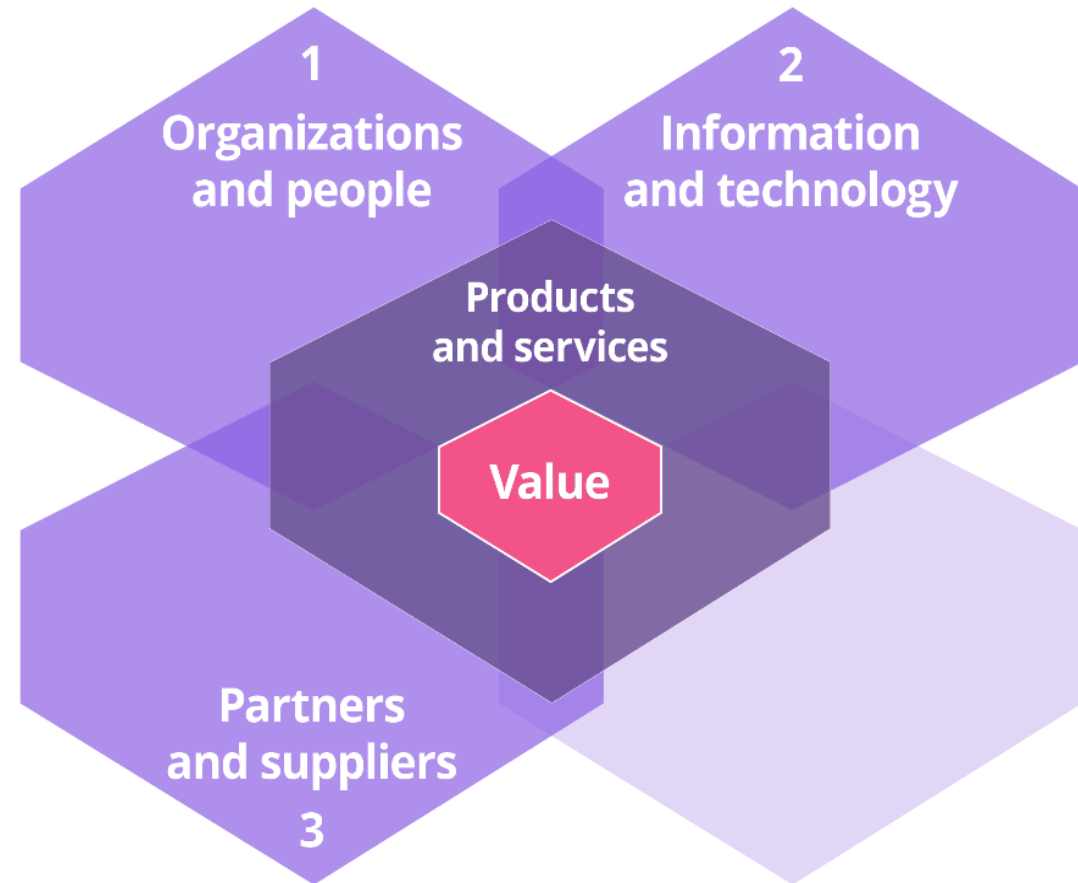
The successful use of automation, data analytics, AI, and digital technology in digital product and service management includes the following three factors:

- **Ensure that the use of technology is considered from the outset** by developing and maintaining an information and technology management approach throughout the product and service lifecycle.
- **Understand and assess** the information and technology capabilities required across the organization's value system, identify capability gaps, and develop action plans to address those gaps.
- **Assess information and technology capability across key areas**, including governance, data, skills, leadership, and infrastructure, to support effective and ethical use of technology and AI.



Partners and suppliers

Every organization depends to some extent on services provided by other organizations.



Adapted from Figure 1.4 The ITIL Four Dimensions of Product and Service management

Working with partners and suppliers

Organizations depend on third-party services, but these relationships vary in scope, formality, and responsibility.

Roles vary:



from providing resources to taking full responsibility for products or services.

Formality differs:



ranges from strict contracts to flexible partnerships.

Managing dependencies:



may involve in-house control or delegated service integration.

Factors influencing supplier strategy

An organization's supplier strategy is shaped by several key factors:

- **Strategic focus**: core versus non-core and outsourced functions.
- **Corporate culture**: historical preferences, bias.
- **Resource scarcity**: lack of needed skills/resources.
- **Cost concerns**: optimizing by whether sourcing externally is more economical.
- **Subject matter expertise**: leveraging supplier knowledge.
- **External constraints**: regulations, policies, codes of conduct.
- **Demand patterns**: seasonal or variable demand.



ITIL Car Rental's partners and suppliers



Max: ITIL Car Rental depends on a network of third-party organizations to deliver high-quality services. Some of these suppliers support us behind the scenes, while others are deeply embedded in our product and service management activities. Together, they help us scale, specialize, and innovate across all areas of our business.



Anna: These partnerships bring a range of considerations. With the manufacturer of our driverless vehicles, deep integration and shared accountability were essential, driven by safety requirements, their subject matter expertise, and reputation. In contrast, for charging services, priorities were different: flexibility and reach were paramount, and the basic contract proved sufficient.

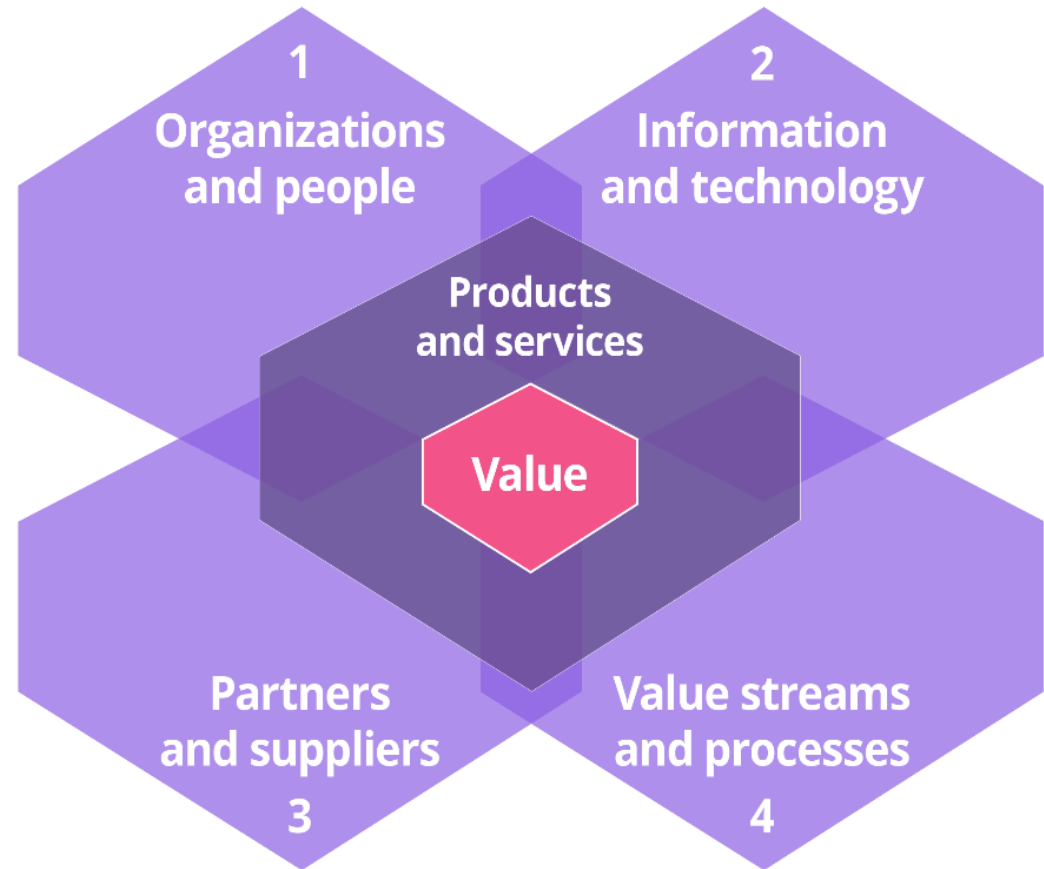


Max: Partner selection criteria are influenced by several factors, including availability of competencies, economic feasibility, strategic relevance, and regulatory compliance. Demand patterns are also considered. For instance, in cities with heavy tourist traffic, we could bring in additional partners to help increase capacity during seasonal peaks.

Value streams and processes

The **value streams and processes** dimension addresses organizational and cross-organizational **workflows**, focusing on what activities the organization undertakes and how they are organized to enable value for stakeholders effectively and efficiently.

Organizations should embrace the **varying complexity** of the **work contexts** and develop the ability to respond according to the specific circumstances.



Adapted from Figure 1.4 The ITIL Four Dimensions of Product and Service management

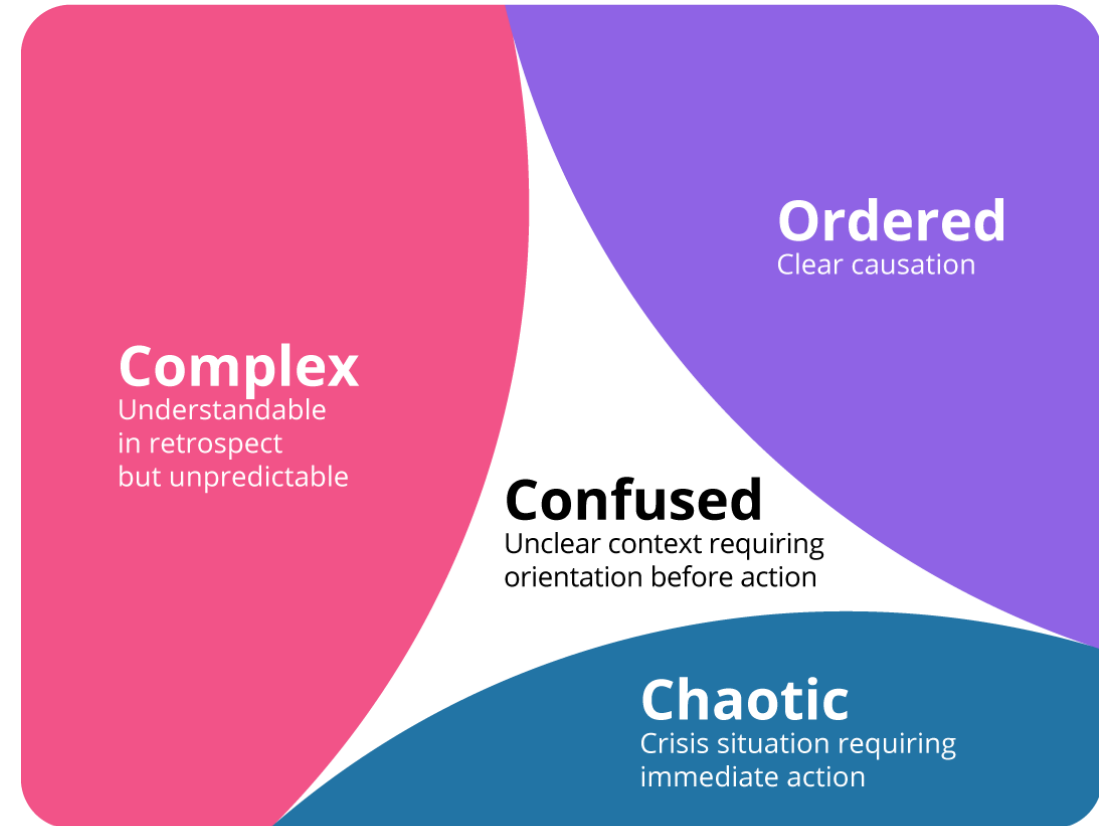
Optimizing workflows for complexity



Complexity thinking

An approach to analysis and decision-making based on the recognition and understanding of the various levels of complexity inherent in the systems and the context in which they operate.

In ITIL, contextual variations are described as four broad categories: **ordered**, **complex**, **chaotic**, and **confused**.



The colours and sizes used in this figure are for illustration purposes only

Figure 2.1 Four types of complexity context